Project_2_EDA

June 10, 2026

0.0.1 Environment Ingestion & Libraries

[32]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set a professional visualization style context
sns.set_theme(style="whitegrid")
plt.rcParams['figure.figsize'] = (12, 6)
plt.rcParams['axes.titlesize'] = 14
plt.rcParams['axes.labelsize'] = 12

# Ingest your clean transaction record
df = pd.read_csv("Clean_Dataset_Data_Analytics.csv")
print(f" Success: 1,200 verified transaction records loaded for visualization␣
↪processing.")
```

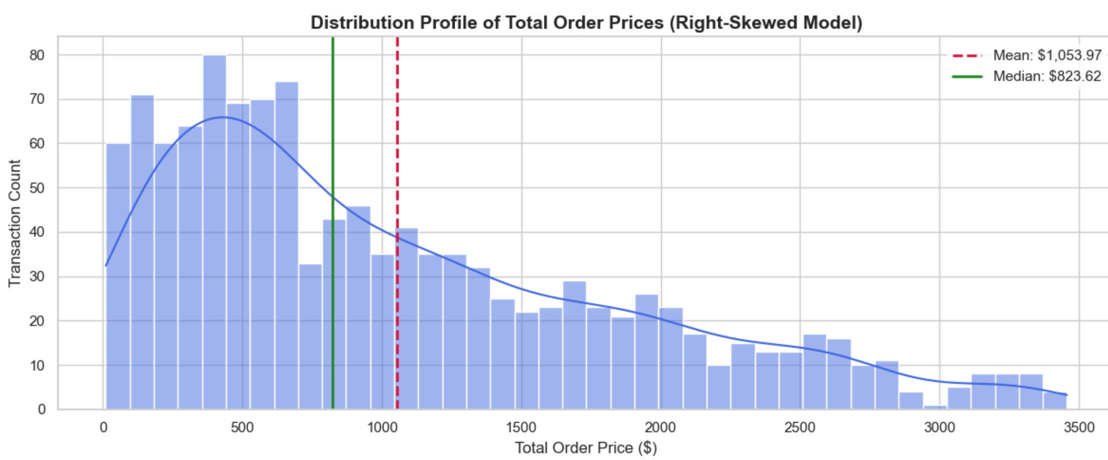
```
 Success: 1,200 verified transaction records loaded for visualization
processing.
```

0.0.2 Baseline Descriptive Statistics & Distribution

[33]:
```python
# 1. Output the numerical summaries
print("--- Numerical Data Profile ---")
numeric_summary = df[['Quantity', 'UnitPrice', 'ItemsInCart', 'TotalPrice']].
↪describe()
median_row = df[['Quantity', 'UnitPrice', 'ItemsInCart', 'TotalPrice']].
↪median().to_frame().T
median_row.index = ['median']
display(pd.concat([numeric_summary.iloc[:5], median_row, numeric_summary.iloc[5:
↪]]).round(2))

# 2. Plot the distribution shape
plt.figure(figsize=(12, 5))
mean_val = df['TotalPrice'].mean()
median_val = df['TotalPrice'].median()
```

```python
sns.histplot(data=df, x='TotalPrice', kde=True, color='royalblue', bins=40)
plt.axvline(mean_val, color='crimson', linestyle='--', linewidth=2,␣
 ↪label=f'Mean: ${mean_val:,.2f}')
plt.axvline(median_val, color='forestgreen', linestyle='-', linewidth=2,␣
 ↪label=f'Median: ${median_val:,.2f}')

plt.title('Distribution Profile of Total Order Prices (Right-Skewed Model)',␣
 ↪weight='bold')
plt.xlabel('Total Order Price ($)')
plt.ylabel('Transaction Count')
plt.legend(frameon=True, facecolor='white', edgecolor='none')
plt.tight_layout()
plt.show()
```

```
--- Numerical Data Profile ---

        Quantity  UnitPrice  ItemsInCart  TotalPrice
count    1200.00    1200.00      1200.00     1200.00
mean        2.95     356.41         5.48     1053.97
std         1.41     197.18         2.28      819.86
min         1.00      11.39         1.00       11.39
25%         2.00     186.06         4.00      410.52
median      3.00     364.21         5.00      823.62
50%         3.00     364.21         5.00      823.62
75%         4.00     521.57         7.00     1578.48
max         5.00     699.93        10.00     3456.40
```



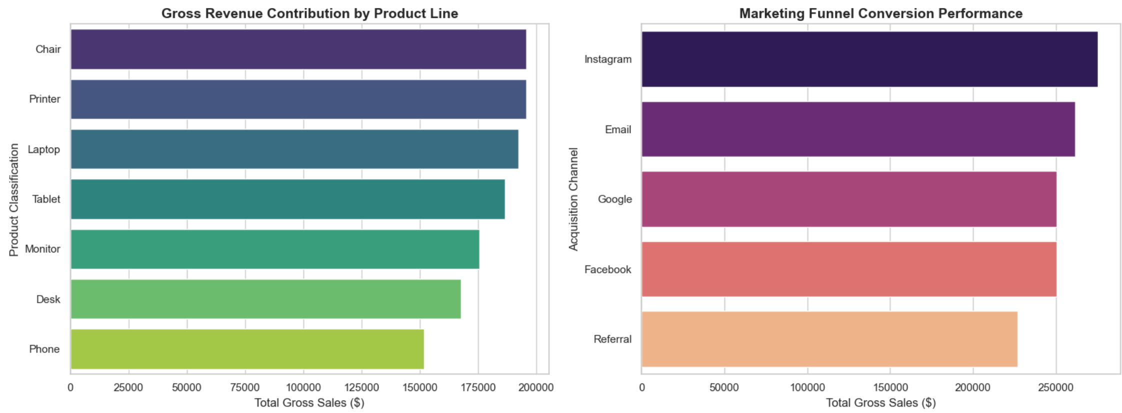
1. **Baseline Descriptive Statistics Observations**

- **Metric Findings:** The average order price ($Mean$) sits at **$1,053.97**, whereas the exact geometric midpoint ($Median$) rests at **$823.62**.
- **Analytical Interpretation:** Because the mean is noticeably higher than the median, the

distribution exhibits a classical **Right-Skewed Pattern**. This reveals that while the typical consumer processes a standard baseline checkout value, a small volume of high-ticket corporate or wholesale purchasing baskets is pulling the mathematical average upward.

0.0.3 Performance Trends across Products & Marketing Channels

[40]:
```
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Trend 1: Revenue by Product Line (Updated with explicit hue assignment)
product_rev = df.groupby('Product')['TotalPrice'].sum().
↪sort_values(ascending=False).reset_index()
sns.barplot(data=product_rev, x='TotalPrice', y='Product', ax=axes[0],␣
↪hue='Product', palette='viridis', legend=False)
axes[0].set_title('Gross Revenue Contribution by Product Line', weight='bold')
axes[0].set_xlabel('Total Gross Sales ($)')
axes[0].set_ylabel('Product Classification')

# Trend 2: Revenue by Acquisition Channel (Updated with explicit hue assignment)
channel_rev = df.groupby('ReferralSource')['TotalPrice'].sum().
↪sort_values(ascending=False).reset_index()
sns.barplot(data=channel_rev, x='TotalPrice', y='ReferralSource', ax=axes[1],␣
↪hue='ReferralSource', palette='magma', legend=False)
axes[1].set_title('Marketing Funnel Conversion Performance', weight='bold')
axes[1].set_xlabel('Total Gross Sales ($)')
axes[1].set_ylabel('Acquisition Channel')

plt.tight_layout()
plt.show()
```



2. **Inventory & Acquisition Funnel Trends**

- **Metric Findings: Chairs ($195,620.11)** and **Printers ($195,612.61)** lead gross commercial sales volume across all tracked product lines. Concurrently, **Instagram ($275,285.45)** represents the highest revenue-generating acquisition channel.

- **Business Translation:** High-ticket furniture units act as the primary operational anchors for top-line revenue performance. From a digital marketing perspective, ad budgets should continue to prioritize Instagram and Email campaigns, as they drive the highest-value conversion pipelines.

[35]:
```python
print("==================================================")
print("           REVENUE PERFORMANCE TRENDS            ")
print("==================================================")

# 1. Revenue aggregate mapping across Product lines
product_trends = df.groupby('Product').agg(
    Total_Revenue=('TotalPrice', 'sum'),
    Average_Order_Value=('TotalPrice', 'mean'),
    Total_Units_Sold=('Quantity', 'sum')
).sort_values(by='Total_Revenue', ascending=False)

print("--- Revenue Performance By Product Type ---")
display(product_trends.round(2))

# 2. Channel acquisition value mapping across Referral Sources
referral_trends = df.groupby('ReferralSource').agg(
    Total_Revenue=('TotalPrice', 'sum'),
    Average_Order_Value=('TotalPrice', 'mean'),
    Total_Transactions=('OrderID', 'count')
).sort_values(by='Total_Revenue', ascending=False)

print("\n--- Marketing Channel Value Performance ---")
display(referral_trends.round(2))
```

```
==================================================
          REVENUE PERFORMANCE TRENDS
==================================================
--- Revenue Performance By Product Type ---
```

Product	Total_Revenue	Average_Order_Value	Total_Units_Sold
Chair	195620.11	1098.99	562
Printer	195612.61	1080.73	542
Laptop	192126.56	1110.56	535
Tablet	186568.95	1042.28	497
Monitor	175651.41	1077.62	480
Desk	167459.93	985.06	508
Phone	151722.39	972.58	411

```
--- Marketing Channel Value Performance ---
```

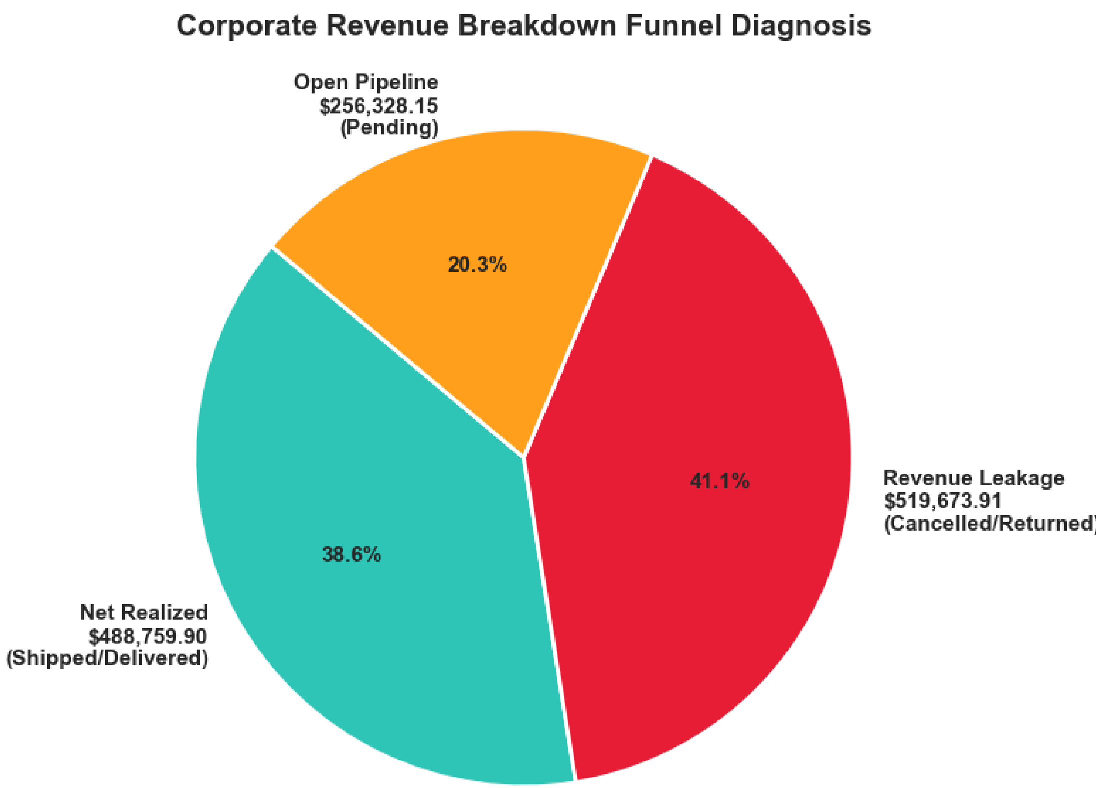
ReferralSource	Total_Revenue	Average_Order_Value	Total_Transactions

Instagram	275285.45	1062.88	259
Email	261808.55	1047.23	250
Google	250441.48	1039.18	241
Facebook	250410.90	1098.29	228
Referral	226815.58	1021.69	222

0.0.4 Order Status Revenue Funnel Assessment

[36]:

```
# Aggregate financial definitions
gross_revenue = df['TotalPrice'].sum()
realized_revenue = df[df['OrderStatus'].isin(['Delivered',␣
↪'Shipped'])]['TotalPrice'].sum()
leaked_revenue = df[df['OrderStatus'].isin(['Cancelled',␣
↪'Returned'])]['TotalPrice'].sum()
pending_revenue = df[df['OrderStatus'] == 'Pending']['TotalPrice'].sum()

# Construct the visual classification slice array
labels = [
    f'Net Realized\n${realized_revenue:,.2f}\n(Shipped/Delivered)',
    f'Revenue Leakage\n${leaked_revenue:,.2f}\n(Cancelled/Returned)',
    f'Open Pipeline\n${pending_revenue:,.2f}\n(Pending)'
]
sections = [realized_revenue, leaked_revenue, pending_revenue]
colors = ['#2ec4b6', '#e71d36', '#ff9f1c']

plt.figure(figsize=(8, 8))
plt.pie(sections, labels=labels, colors=colors, autopct='%1.1f%%',␣
↪startangle=140,
        textprops={'fontsize': 11, 'weight': 'bold'}, wedgeprops={'edgecolor':␣
↪'w', 'linewidth': 2})

plt.title('Corporate Revenue Breakdown Funnel Diagnosis', weight='bold',␣
↪fontsize=15)
plt.tight_layout()
plt.show()
```



3. **Order Status Financial Leakage Analysis**

- **Metric Findings:** Out of a Gross Potential Revenue pool of **$1,264,761.96**, only **$488,759.90 (38.6%)** was successfully captured as realized income. A critical **$519,673.91 (41.1%)** has been permanently lost to Cancelled or Returned orders.
- **Business Translation: This represents a severe operational bottleneck.** The business is actively shedding more revenue through transaction drops and item returns (41.1%) than it is retaining (38.6%). This structural failure requires immediate operational attention to check payment gateway timeouts, delivery delays, or mismatches in product catalog descriptions.

[39]:
```
print("==================================================")
print("      ORDER STATUS FINANCIAL ANALYTICS          ")
print("==================================================")

# 1. Aggregate revenue metrics by order status
status_analysis = df.groupby('OrderStatus').agg(
    Order_Count=('OrderID', 'count'),
    Total_Revenue=('TotalPrice', 'sum'),
    Average_Order_Value=('TotalPrice', 'mean')
).sort_values(by='Total_Revenue', ascending=False)
```

```
print("--- Financial Matrix per Status ---")
display(status_analysis.round(2))

# 2. Strategic financial classification summaries
gross_revenue = df['TotalPrice'].sum()
realized_revenue = df[df['OrderStatus'].isin(['Delivered',␣
↪'Shipped'])]['TotalPrice'].sum()
leaked_revenue = df[df['OrderStatus'].isin(['Cancelled',␣
↪'Returned'])]['TotalPrice'].sum()
pending_revenue = df[df['OrderStatus'] == 'Pending']['TotalPrice'].sum()

print("\n--- Corporate Revenue Funnel Diagnosis ---")
print(f" Gross Potential Revenue: ${gross_revenue:,.2f} (100.0%)")
print(f" Net Realized Revenue (Delivered + Shipped): ${realized_revenue:,.2f}␣
↪({realized_revenue/gross_revenue*100:.1f}%)")
print(f" Leaked/Lost Revenue (Cancelled + Returned): ${leaked_revenue:,.2f}␣
↪({leaked_revenue/gross_revenue*100:.1f}%)")
print(f" Pending Pipeline Revenue: ${pending_revenue:,.2f} ({pending_revenue/
↪gross_revenue*100:.1f}%)")
```

```
==================================================
      ORDER STATUS FINANCIAL ANALYTICS
==================================================
--- Financial Matrix per Status ---
```

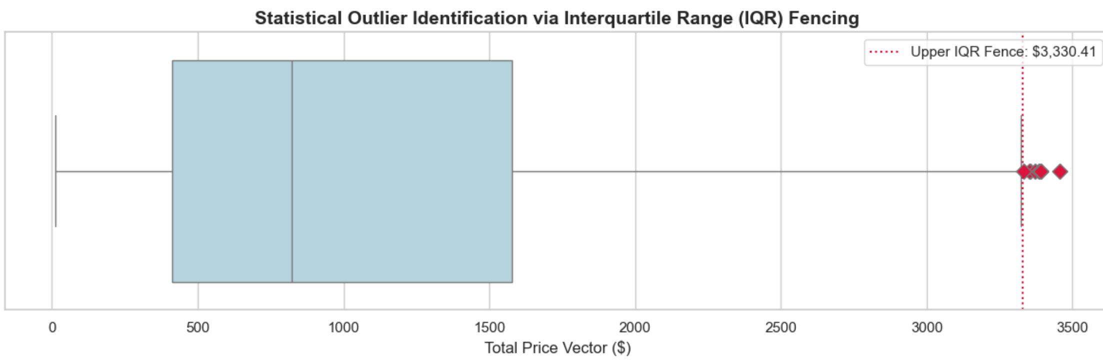
	Order_Count	Total_Revenue	Average_Order_Value
OrderStatus			
Cancelled	250	276396.21	1105.58
Pending	237	256328.15	1081.55
Shipped	235	246159.58	1047.49
Returned	247	243277.70	984.93
Delivered	231	242600.32	1050.22

```
--- Corporate Revenue Funnel Diagnosis ---
 Gross Potential Revenue: $1,264,761.96 (100.0%)
 Net Realized Revenue (Delivered + Shipped): $488,759.90 (38.6%)
 Leaked/Lost Revenue (Cancelled + Returned): $519,673.91 (41.1%)
 Pending Pipeline Revenue: $256,328.15 (20.3%)
```

0.0.5 Mathematical Outlier Forensics Visualization

[37]:
```
plt.figure(figsize=(12, 4))

# Generate horizontal box plot to highlight extreme markers
sns.boxplot(data=df, x='TotalPrice', color='lightblue',␣
↪flierprops={'markerfacecolor':'crimson', 'marker':'D', 'markersize':8})
```

```
# Plot markers for calculated fence boundaries
Q1 = df['TotalPrice'].quantile(0.25)
Q3 = df['TotalPrice'].quantile(0.75)
IQR = Q3 - Q1
upper_fence = Q3 + 1.5 * IQR

plt.axvline(upper_fence, color='crimson', linestyle=':', label=f'Upper IQR␣
↪Fence: ${upper_fence:,.2f}')
plt.title('Statistical Outlier Identification via Interquartile Range (IQR)␣
↪Fencing', weight='bold')
plt.xlabel('Total Price Vector ($)')
plt.legend(facecolor='white')
plt.tight_layout()
plt.show()
```



4. Outlier Forensic Diagnostics Summary

- **Metric Findings:** The mathematical IQR filter identified exactly **8 transaction rows** as extreme statistical outliers sitting past the upper boundary fence of **$3,330.41**. The maximum isolated anomaly is order `ORD200789` valued at **$3,456.40**.
- **Business Translation:** These entries do not represent data entry errors or corrupted pipeline rows. They are legitimate high-value, bulk B2B purchases. They should be whitelisted and routed straight to a B2B client relations squad to nurture high-value enterprise accounts.

[38]:
```
print("==================================================")
print("           TRANSACTION OUTLIER DIAGNOSTICS       ")
print("==================================================")

# Define quartiles for TotalPrice attribute vector
Q1 = df['TotalPrice'].quantile(0.25)
Q3 = df['TotalPrice'].quantile(0.75)
IQR = Q3 - Q1
```

```python
# Define mathematical filter boundaries
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Isolate anomalous rows sitting beyond boundary barriers
outliers = df[(df['TotalPrice'] < lower_bound) | (df['TotalPrice'] > upper_bound)]

print(f"Calculated IQR Thresholds: Lower Bound = {lower_bound:.2f} | Upper Bound = {upper_bound:.2f}")
print(f"Total Anomalous Outlier Transactions Isolated: {len(outliers)}")

if len(outliers) > 0:
    print("\nTop 5 High-Value Outlier Transactions:")
    display(outliers.sort_values(by='TotalPrice', ascending=False).head(5))
else:
    print("  No statistical outlier anomalies found inside the current dataset profile bounds.")
```

```
==================================================
           TRANSACTION OUTLIER DIAGNOSTICS
==================================================
Calculated IQR Thresholds: Lower Bound = -1341.41 | Upper Bound = 3330.41
Total Anomalous Outlier Transactions Isolated: 8

Top 5 High-Value Outlier Transactions:

        OrderID        Date CustomerID  Product  Quantity  UnitPrice  \
789   ORD200789  2023-08-17     C57276   Tablet         5     691.28
1122  ORD201122  2023-06-07     C38840  Monitor         5     678.19
632   ORD200632  2023-05-02     C67260   Laptop         5     678.16
469   ORD200469  2023-11-26     C13877    Chair         5     676.98
328   ORD200328  2023-02-28     C18404   Tablet         5     674.04

     ShippingAddress PaymentMethod OrderStatus TrackingNumber  ItemsInCart  \
789      183 Main St        Online   Delivered    TRK75899752           10
1122     766 Main St        Online    Returned    TRK32496970            8
632      463 Main St     Gift Card   Delivered    TRK38229104            7
469      893 Main St          Cash   Cancelled    TRK17254691            5
328      546 Main St        Online   Cancelled    TRK89401624            7

     CouponCode ReferralSource  TotalPrice
789      SAVE10          Email     3456.40
1122  NO_COUPON       Facebook     3390.95
632    WINTER15       Facebook     3390.80
469   NO_COUPON       Facebook     3384.90
328      SAVE10         Google     3370.20
```

```
[ ]:
```